Graph U-Net With Topology-Feature Awareness Pooling for Hyperspectral Image Classification

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***Abstract*— Nowadays, various graph convolutional networks (GCNs) to process graph-structured data have been proposed for hyperspectral image (HSI) classification. Nevertheless, most GCN-based HSI classification methods emphasize graph node feature aggregation instead of graph pooling, resulting in them being shallow networks and unable to extract deep discriminative features. Besides, to obtain the new graph after the pooling layer, current graph pooling methods used for HSI classification just consider node feature information to select important nodes and directly discard unselected nodes, which could be a subjective process and may cause information loss. To solve this issue, we propose a novel graph U-Net with topology-feature awareness pooling (the so-called TFAP graph U-Net) for HSI classification considering a deep network to extract compelling features and automatically selecting nodes beneficial to classification. More specifically, to establish a more precise pooled graph, the graph's topology structure and node feature information are taken into account, making the node selection process more convincing and objective. Furthermore, to allow that graph nodes preserve more useful graph information, our method aggregates node features from neighboring nodes that may not be selected, which can alleviate the loss of information during the pooling process. Moreover, a cross-attention module (CAM) is used to filter out irrelevant or noisy features. Finally, we evaluate the proposed method on three public HSI datasets, i.e., Indian Pines, University of Pavia (PaviaU), and University of Houston. The experimental results demonstrate the superiority of the proposed approach compared with other state-of-the-art methods.**

***Index Terms*— Classification, graph convolutional network (GCN), graph U-Net, hyperspectral imaging, multiresolution analysis, remote sensing, superpixel feature learning.**

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I. Introduction

HYPERSPECTRAL images (HSIs) allow fine-grained object classification on the ground. Therefore, enhancing the capacity of spectral–spatial feature extraction leveraging on deep features from HSIs is a hot research topic [1], [2]. HSI classification has been widely used in several research fields, such as precision agriculture, medical diagnosis, and geological exploration [3], [4], [5].

In early studies, the majority of HSI classification approaches modeled spectral information based on traditional machine learning, such as support vector machine (SVM) [6], [7], K-nearest neighbor classifier [8], decision tree [9], and logistic regression (LR) [10]. Since hyperspectral sensors enable the acquisition of high-dimensional spectra, traditional machine methods using statistical learning showed a reduced generalization ability for HSI classification limiting their applicability. To solve this problem, some representation-based approaches have been proposed to extract correlation information among spectral bands, such as low-rank representation [11], [12] and sparse representation [13], [14]. However, these methods need expert knowledge and handcrafted features. To tackle this drawback, deep-learning-based methods, as a powerful tool in feature extraction, have been used to extract discriminative features from HSIs without handcrafted feature engineering. Convolutional neural networks (CNNs) [15] are the backbone of most deep-learning-based approaches, benefiting from their powerful capacity to extract spectral–spatial features from HSIs. The first CNN-based method used for HSI classification has been presented in [16], demonstrating its superior effectiveness with respect to traditional approaches. Subsequently, a two-stream CNN method [17] has been proposed to make full use of local–global spectral–spatial features. Besides, the densely connected CNN [18] and the residual connected CNN [19], [20] have been proposed to build deeper architectures. Apart from CNNs, other deep-learning-based methods have also achieved good performance in HSI classification, such as recurrent neural networks (RNNs) [21], [22], [23], capsule networks [24], [25], [26], generative adversarial networks (GANs) [27], [28], [29], deformable convolutional networks (DCNs) [30], [31], stacked autoencoders (SAEs) [32], [33], and vision transformer (ViT)-based networks [34], [35], [36], [37]. RNN-based methods mainly exploit spectral information (i.e., the correlation among spectral bands). Hu et al. [38] presented a convolutional long short-term memory (ConvLSTM) network to extract discriminative

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spectral–spatial features from HSIs, which extended LSTM to explore the structure information of HSIs. However, for CNN-based methods, overfitting is likely to occur with a small number of training samples. Another limitation is that CNNs process Euclidean data, which means that they are exploited for handling data with regular spatial structures, greatly restricting application scenarios.

Recently, graph convolutional networks (GCNs) have been widely used in HSI classification, thanks to their robust feature representation ability [39]. Unlike CNN-based methods, GCNs can represent non-Euclidean data (such as graph structure data) by aggregating neighbor node information [40], [41], [42]. In the domain of HSI classification, a semi-supervised GCN [43] has been proposed to exploit neighbor nodes to approximate the convolution operation. However, treating each pixel as a graph node can result in a high computational burden. To solve this problem, the mini-batch GCN [44] has been presented by transforming the original graph into small patches during the training process. In addition, some superpixel-based methods [45], [46], [47] have been introduced to reduce the computation burden taking each superpixel as a node. To extract features at pixel level and boundaries among different semantic classes, Liu et al. [48] presented a CNN-enhanced GCN, which can simultaneously extract features at superpixel and pixel levels leveraging on both GCNs and CNNs. Nevertheless, owing to the oversmooth problem, most GCN-based HSI classification methods use (two or three) shallow layers, significantly limiting the ability to extract discriminative features from HSIs. To solve this drawback, Yu et al. [49] designed a two-branch GCN to extract spectral-spatial features from HSIs, which uses residual connection and dropout strategy to alleviate overfitting and oversmoothing phenomena. Furthermore, Guo et al. [50] introduced a two-stream graph U-Net with pooling and unpooling (Unpool) methods to build a deep network. Besides, Liu et al. [51] proposed a multilevel superpixel structured graph U-Net (MSSGU) to extract several multiscale discriminative features through building hierarchical segmentation graphs. Moreover, Chen et al. [52] presented an offset graph U-Net (OGU) by building multifeature adjacency matrix (spectral, spatial, and context-aware) to fully utilization superpixel-level information. Despite their success, these graph U-Net-based methods still face with some challenges. Indeed, during the pooling process, these methods only rely on node features to select important nodes, thus ignoring the structural information of the entire graph. Moreover, unselected nodes are directly discarded during the graph pooling process, inevitably resulting in a substantial loss of graph feature information. Finally, to fuse the advantages of different stages of U-Net, these methods just use skip connection or summation to combine different scale graph features without filtering useless (or noisy) features.

To solve the above-mentioned problems, a novel graph U-Net with topology-feature awareness pooling (the so-called TFAP graph U-Net) is proposed for HSI classification. Unlike the existing graph U-Net based HSI classification methods that just consider node features in obtaining the pooled graph topology, we simultaneously consider topology structure and feature information of the graph to select important graph nodes during the pooling process. To ensure that nodes after pooling contain more graph information, the selected node features aggregate features from neighboring graph nodes that could be unselected, which can ensure that significant features are retained to a certain extent, thereby enhancing the node's representational capacity. Besides, benefiting from graph pooling and Unpool layers, on one hand, we can build a deep network to extract more discriminative features and alleviate oversmoothing issues; on the other hand, the network can learn multiresolution features from the input image. Furthermore, we introduce a cross-attention module (CAM) to fuse different stage features, thus filtering useless (or noisy) features. More specifically, we convert the original HSI into superpixel-level representation via a superpixel map. Afterward, the TFAP graph U-Net is exploited to extract discriminative spectral–spatial features. Finally, we convert the superpixel-level representation into the related pixel-level representation to get the final classification map.

The main contributions of this work can be summarized as follows.

1) We propose a graph U-Net with TFAP for HSI classification that simultaneously uses topological structure and feature information to select important nodes during the graph pooling process.
2) Unlike the existing graph U-Net-based HSI classification methods that directly discard unselected graph nodes when generating a new graph after the pooling layer, the proposed TFAP graph U-Net aggregates unselected node information before discarding nodes, which can improve the use of feature information from unselected nodes.
3) To fully fuse multiresolution features extracted by graph U-Nets, we propose a CAM to filter out useless (or noisy) areas from skip connection features.

The remaining of this article is organized as follows. Section II describes the proposed method. A broad experimental analysis on three HSI datasets is shown in Section III. Finally, concluding remarks are drawn in Section IV.

II. Proposed Approach

The overall architecture of the proposed method is depicted in Fig. 1. We first need to segment the HSI in input into superpixel-level regions. Afterward, the TFAP graph U-Net is designed for extracting graph features. Finally, the extracted graph features are transformed into pixel-level to get the pixel-level classification.

A. Notation

The mathematical notation used in this article is as follows. Scalars are denoted by lowercase letters (e.g., x). Matrices (2-D and 3-D arrays) and vectors are expressed in uppercase bold letters (e.g., $\mathbf{X}$) and lowercase bold letters (e.g., $\mathbf{x}$), respectively. Euler script letters are used for sets (e.g., $\mathcal{G}$). We define the input HSI as $\tilde{\mathbf{X}} \in \mathbb{R}^{H\times W\times B}$ with H, W, and B being the height, width, and the number of spectral bands, respectively. $\mathbf{X} = \{\mathbf{x}_1, \mathbf{x}_2, \ldots, \mathbf{x}_M\}$ is the graph representation of $\mathcal{G}$, where M is the number of nodes (superpixels). Finally, $\mathbf{x}_i$ refers to the ith node.

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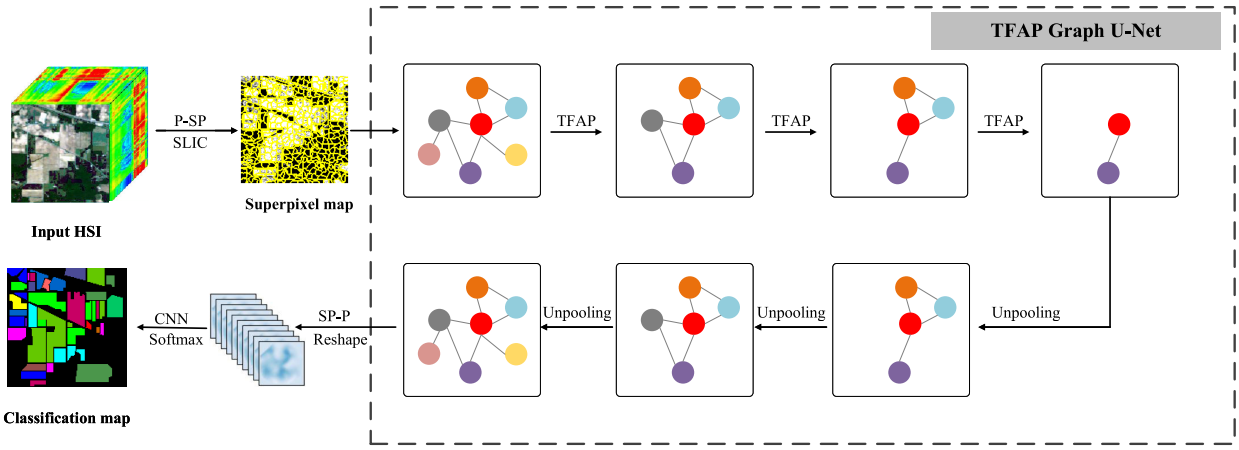


Fig. 1. Architecture of the proposed method.

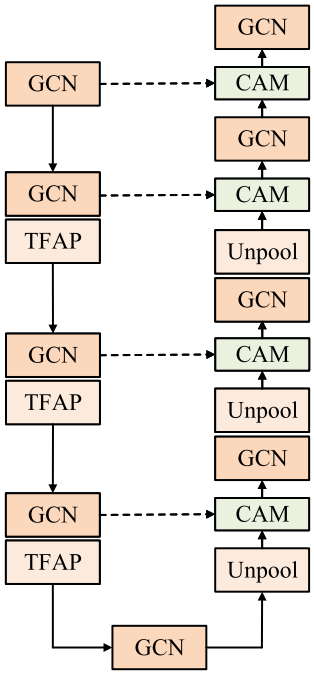


Fig. 2. Illustration of the TFAP graph U-Net module. The dotted line refers to skip connection.

B. Superpixel Map

Considering that pixel-level graphs lead to a high computational burden, we use the simple linear iterative clustering (SLIC) [53] to segment the HSI in input into some small local regions (i.e., superpixels), each of which contains pixels with a strong spectral–spatial similarity. Each superpixel is treated as a graph node. More specifically, the HSI, $\tilde{\mathbf{X}} \in \mathbb{R}^{H\times W\times B}$, is divided into $M = \lceil (H \times W)/\delta \rceil$ superpixels, where $\lceil\cdot\rceil$ is the ceiling operator and δ is a parameter to control the number of superpixels in the image. After getting the superpixel map, we can define the assignment matrix, $\mathbf{O} \in \mathbb{R}^{HW\times M}$, among pixels and superpixels

$$\mathbf{O}_{i,j} = \begin{cases} 1, & \text{if } \tilde{\mathbf{X}}_i \in S_j \\ 0, & \text{otherwise} \end{cases} \tag{1}$$

where $\mathbf{O}_{i,j}$ is the ith pixel belonging to the jth superpixel, $\tilde{\mathbf{X}}_i$ denotes the ith pixel in $\tilde{\mathbf{X}} \in \mathbb{R}^{HW\times B}$, $\bar{\mathbf{X}} = \text{Flatten}(\tilde{\mathbf{X}})$, $\text{Flatten}(\cdot)$ is the flatten operator, and S_j indicates the jth superpixel.

C. TFAP Graph U-Net

Graph convolution network (GCN) [54], a multilayer neural network, is a robust tool designed for managing data organized in graph structures, which can aggregate node features from its neighbors and can propagate information via edges, exhibiting encouraging performance in a variety of tasks. In this work, we use GCNs to extract spectral–spatial features from HSIs, which can be defined as

$$\mathbf{H}^{(l+1)} = \sigma\left(\widetilde{\mathbf{D}}^{-\frac{1}{2}}\widetilde{\mathbf{A}}\widetilde{\mathbf{D}}^{-\frac{1}{2}}\mathbf{H}^{(l)}\mathbf{W}^{(l+1)}\right) \tag{2}$$

where $\mathbf{H}^{(l+1)}$ is the output of the $(l+1)$th layer, $\sigma(\cdot)$ denotes the activation function, and $\mathbf{W}^{(l+1)}$ represents a trainable parameter. $\widetilde{\mathbf{A}} = \mathbf{A} + \mathbf{I}$ and $\widetilde{\mathbf{D}}_{ii} = \sum_j \widetilde{\mathbf{A}}_{ij}$ are the adjacency matrix with self-loops and the degree matrix of $\widetilde{\mathbf{A}}$, respectively, and $\mathbf{I}$ is an identity matrix. The adjacency matrix, $\mathbf{A}$, is based on the spectral similarity distance calculated as

$$\mathbf{A}_{ij} = \begin{cases} e^{-\gamma\|(\mathbf{x}_i-\mathbf{x}_j)\|^2}, & \text{if } \mathbf{x}_i \in \mathrm{N}(\mathbf{x}_j) \text{ or } \mathbf{x}_j \in \mathrm{N}(\mathbf{x}_i) \\ 0, & \text{otherwise} \end{cases} \tag{3}$$

where γ is the weight of the edges, $\mathbf{x}_i$ and $\mathbf{x}_j$ denote two graph nodes, and $\mathrm{N}(\mathbf{x})$ represents the neighbors of $\mathbf{x}$.

The basic graph U-Net [55] contains the graph pooling layer and the graph Unpool layer, which can dynamically select a subset of graph nodes from the raw graph to obtain a smaller graph, thus increasing the extraction feature capabilities. A key issue for graph U-Net is how to design a more effective pooling strategy to preserve most of the informative graph nodes. In this section, we will present the TFAP graph U-Net, which is shown in Fig. 2. Two GCN layers are used to extract shallow features. Afterward, several blocks of GCN and TFAP layers are considered. The decoder branch shows the same number of blocks, but containing a GCN layer, an Unpool layer, and a CAM to further extract discriminative features.

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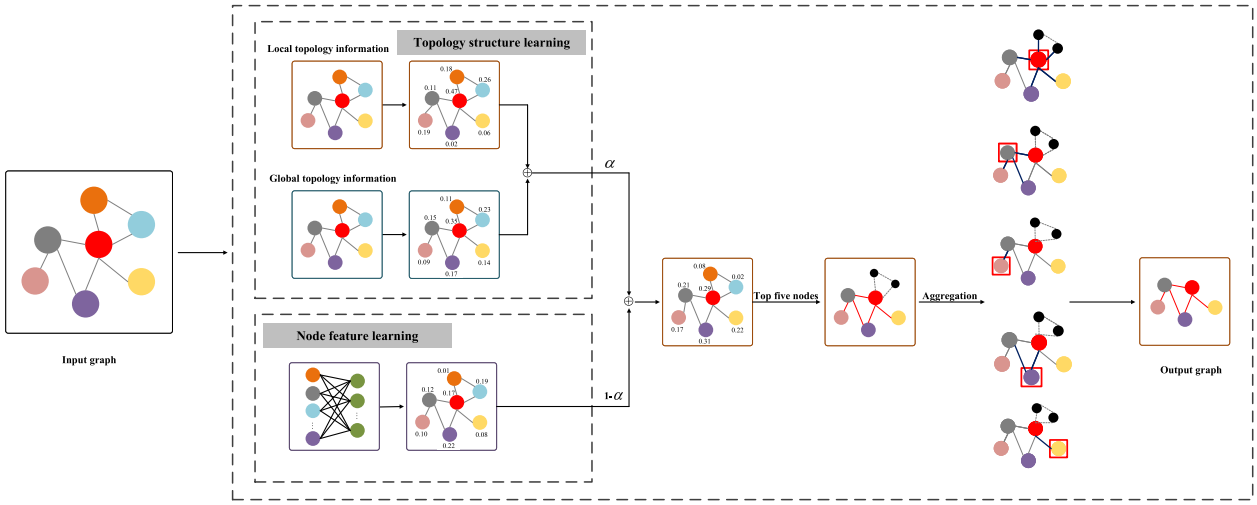


Fig. 3. Illustration of the TFAP layer. ⊕ indicates the summation operator. The black circles denote the unselected nodes.

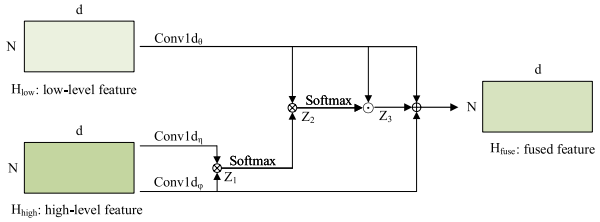


Fig. 4. Illustration of the CAM. The Hadamard product operation is indicated with ⊙, instead, ⊕ and ⊗ denote the summation and multiplication operators, respectively.

1) Topology-Feature Awareness Pooling: Pooling layers are used between consecutive convolutional layers to decrease data volume and parameters, thereby preventing overfitting. The core idea of sampling-based graph pooling lies in establishing a criterion that yields a ranking assessing the significance of nodes to obtain a smaller graph. To fully use both topology structure and node feature information, we propose the TFAP (see Fig. 3). We first conduct topology structure and node feature learning to obtain the fusion score by the hyperparameter α. Then, several important graph nodes are selected considering the fusion score. Finally, the neighbor information of each selected node is aggregated.

a) Topology structure learning: Generally speaking, a graph with a large number of nodes and edges retains a rich structural information. Hence, it is necessary to select important nodes according to their topological structure information during the pooling process. We consider local and global topology information to select the subset of graph nodes. Regarding local topology information, we assume that if a target node is involved in reconstructing its neighborhood graph nodes, it means that the corresponding node holds significant importance. That is, local topology information measures the importance of target node within its neighborhood [56]. In addition, if a target node has a large degree (i.e., with many neighbors containing abundant information), this latter is also considered an important one [57]. Hence, the local topology information score contains two parts. One is the Manhattan distance between the target node and the reconstructed one from its neighbors; the other one is the degree information of the target node

$$\mathbf{p}_{lt1}^{l} = \gamma(\mathcal{G}_i) = \| \left(\mathbf{I}_i^l - \left(\mathbf{D}_i^l \right)^{-1} \mathbf{A}_i^l \right) \mathbf{H}_i^l \|_1 \tag{4}$$

$$\mathbf{p}_{lt2}^{l} = \text{sigmoid}\left(\sum_i \mathbf{A}_{i:}^l \right) \tag{5}$$

$$\mathbf{p}_{lt}^{l} = \mathbf{p}_{lt1}^{l} + \mathbf{p}_{lt2}^{l} \tag{6}$$

where $\mathbf{p}_{lt}^{l}$ represents the lth layer local topology information score, $\| \cdot \|_1$ indicates the ℓ_1-norm rowwisely, $\mathbf{p}_{lt1}^{l} \in \mathbb{R}^{n_i}$ with n_i being the number of nodes in the ith graph, $\mathbf{A}^l$ is the lth layer adjacency matrix with self-loops, with $\mathbf{D}^l$ denoting the related degree matrix, and $\mathbf{H}^l$ refers to the lth layer node representation.

Regarding global topology information, we use a GCN to explore the graph structure information, which is more suitable for measuring the importance of graph nodes [58], [59]. The global topology information denotes the importance of different local regions in the entire graph

$$\mathbf{p}_{gt}^{l} = \text{softmax}\left(\left(\mathbf{D}^l \right)^{-1} \mathbf{A}^l \mathbf{H}^l \mathbf{W}^l \right) \tag{7}$$

where $\mathbf{p}_{gt}^{l}$ is the lth layer global topology information score and $\mathbf{W}^l$ is a trainable parameter. Consequently, the topology structure information score, $\mathbf{p}_t^l$, is calculated as follows:

$$\mathbf{p}_t^l = \mathbf{p}_{lt}^{l} + \mathbf{p}_{gt}^{l}. \tag{8}$$

b) Node feature learning: In addition to topological information, it is essential to use node feature information in the pooling process because graph nodes contain important feature information and node features can roughly describe object attributes. To this aim, multiple-layer perceptron (MLP) [60] is used to extract graph node features

$$\mathbf{p}_f^l = \sigma\left(\text{MLP}\left(\mathbf{H}^l \right) \right) \tag{9}$$

where $\mathbf{p}_f^l$ is the node feature information score, and σ represents an activation function.

To enhance the nodes' assessment, graph topology structure and node feature information are exploited to calculate node scores

$$\mathbf{s}^l = \lambda \mathbf{p}_t^l + (1 - \lambda) \mathbf{p}_f^l \tag{10}$$

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where $\mathbf{s}^l$ is the lth layer node score, and λ is a hyperparameter.

After obtaining the final node important score, downsampling will be applied to the graph data. Top k nodes (based on node scores) are dynamically selected from the current graph nodes. TFAP pooling aggregates the information from the neighbors of each selected node that may be unselected. In this way, even if pivotal nodes are removed, the feature aggregation process will retain crucial information to a certain degree. The pooling process can be expressed as

$$idx = \text{rank}\left(\mathbf{s}^l, k\right) \tag{11}$$

$$\mathbf{H}^{\text{att}^l} = \mathbf{s}^l \odot \mathbf{H}^l \tag{12}$$

where rank$(\cdot)$ denotes the ranking function, idx is the index of the selected node, and $\odot$ is the Hadamard product. $\mathbf{H}^{\text{att}^l} \in \mathbb{R}^{n_1 \times c_1}$ is the weighted graph representation, where n_1 and c_1 denote the number of nodes and the node dimension, respectively. According to idx, we can obtain the subgraph containing important parts, whose adjacency matrix is defined as

$$\hat{\mathbf{A}} = \mathbf{A}^l_{\text{idx}(i)} \tag{13}$$

where $\hat{\mathbf{A}} \in \mathbb{R}^{k \times n_1}$. Subsequently, for the selected nodes, the new feature representation $\mathbf{H}^{l+1}$ is pooled from $\mathbf{H}^{\text{att}^l}$, which can aggregate from the neighbors of each selected node

$$\mathbf{H}^{l+1} = \hat{\mathbf{A}}\mathbf{H}^{\text{att}^l} \tag{14}$$

$$\mathbf{A}^{l+1} = \mathbf{A}^l(\text{idx}, \text{idx}) \tag{15}$$

where $\mathbf{H}^{l+1} \in \mathbb{R}^{k \times c_1}$, and $\mathbf{A}^{l+1} \in \mathbb{R}^{k \times k}$ is the new adjacency matrix of the pooled graph.

2) Unpool: The goal of TFAP is to select a subset of nodes in the graph to reduce the number of nodes. Conversely, in the Unpool layer, we need to upsample the nodes in the graph to restore them to their original topology structure. To achieve it, based on the positions of the selected nodes stored in the corresponding TFAP layer, we relocate the nodes to their original positions in the graph. The Unpool process is defined as follows:

$$\mathbf{H}^{l+1} = \text{distribute}\left(\mathbf{0}, \mathbf{H}^l, \text{idx}\right) \tag{16}$$

where idx is the index of the nodes in the corresponding TFAP layer, $\mathbf{0}$ denotes an all-zeros matrix that is the initial graph representation for the new graph, and distribute$(\cdot)$ arranges (based on the indexes specified in idx) the row vectors in $\mathbf{H}^l$ into the graph representation $\mathbf{0}$.

3) CAM: Thanks to the TFAP and the Unpool layers, the graph U-Net structure can extract multistage spatial–spectral features that is different from nonlocal neural networks [61], which focus on capturing long-distance dependencies and global information by calculating interactions between positions of arbitrary features. In this work, we propose a CAM to enhance the low-level features between TFAP–Unpool pairs. On one hand, it can prevent the loss of feature information as the number of network layers deepens. On the other hand, the proposed CAM can filter out the irrelevant (or noisy) features through the residual connection highlighting features useful for HSI classification. The illustration of the CAM is depicted in Fig. 4. The low-level and high-level features having the same size are extracted by the TFAP layer and the corresponding Unpool. First, we apply three 1×1 convolution kernels to extract features

$$\mathbf{Z}_1 = \sigma\left(\left(w_\eta \mathbf{H}_{\text{high}}\right)\left(w_\psi \mathbf{H}_{\text{high}}\right)^T\right) \tag{17}$$

where $\mathbf{Z}_1 \in \mathbb{R}^{N \times N}$ denotes the relationships between nodes in higher order graph, $\mathbf{H}_{\text{high}} \in \mathbb{R}^{N \times d}$, σ is the softmax function, and w_η and w_ψ are the convolution kernels

$$\mathbf{Z}_2 = \sigma\left(\mathbf{Z}_1(w_\theta \mathbf{H}_{\text{low}})\right) \tag{18}$$

where $\mathbf{Z}_2 \in \mathbb{R}^{N \times d}$ denotes the relationships between nodes in higher order graphs and lower order graphs, which can emphasize nodes in low-level graph that are important for HSI classification. $\mathbf{H}_{\text{low}} \in \mathbb{R}^{N \times d}$ and w_θ is the convolution kernel. We use the Hadamard product $(\odot)$ to filter out irrelevant or noisy information from low-level graph representation

$$\mathbf{Z}_3 = \mathbf{Z}_2 \odot (w_\theta \mathbf{H}_{\text{low}}) \tag{19}$$

where $\mathbf{Z}_3 \in \mathbb{R}^{N \times d}$ denotes the graph representation of being cleaned. Finally, we use the summation operation $(\oplus)$ to fuse the low-level and high-level features

$$\mathbf{H}_{\text{fused}} = \mathbf{Z}_3 + (w_\theta \mathbf{H}_{\text{low}}) + \left(w_\psi \mathbf{H}_{\text{high}}\right) \tag{20}$$

where $\mathbf{H}_{\text{fused}} \in \mathbb{R}^{N \times d}$ denotes the fused features.

D. Classification

After obtaining the final graph node representation, $\mathbf{H}_{\text{final}}$, extracted by the TFAP graph U-Net, we need to project it into pixel level (SP-P) based on the assignment matrix, $\mathbf{O}$. Thus, we have that

$$\mathbf{P} = \text{Reshape}(\mathbf{O}\mathbf{H}_{\text{final}}) \tag{21}$$

where Reshape$(\cdot)$ refers to the reconstruction of the spatial dimensions of the original data. To further explore discriminative features, the pixel-level features, $\mathbf{P}$, are fed into 2-D convolution layers. Finally, a softmax layer is used for predicting per-pixel class probabilities. In this work, the classic cross-entropy error is adopted as HSI classification loss, $\mathcal{L}$, which is defined as

$$\mathcal{L} = -\sum_{i=1}^{T}\sum_{j=1}^{C} \mathbf{Y}_{ij} \log\left(\hat{\mathbf{Y}}_{ij}\right) \tag{22}$$

where T and C are the number of training samples and classes, respectively, log is the natural logarithm, while $\mathbf{Y}$ and $\hat{\mathbf{Y}}$ denote the (ground-truth) label and predicted matrices, respectively.

III. Experimental Analysis

To evaluate the effectiveness of the proposed method, we conducted extensive experiments on three public HSI datasets, i.e., Indian Pines, University of Pavia (PaviaU), and Houston University (Houston). Four evaluation metrics, i.e., per-class accuracy, overall accuracy (OA), average accuracy (AA), and Kappa coefficient (Kappa), have been used to evaluate the performance of the compared approaches. The code is available at https://github.com/sincos-cr/graph-pooling.git

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TABLE I
STATISTICS OF THE INDIAN PINES DATASET

Class No.	Class Name	Training	Testing
1	Alfalfa	30	16
2	Corn-notill	30	1398
3	Corn-mintill	30	800
4	Corn	30	207
5	Grass-pasture	30	453
6	Grass-trees	30	700
7	Grass-pasture-mowed	15	13
8	Hay-windrowed	30	448
9	Oats	15	5
10	Soybean-notill	30	942
11	Soybean-mintill	30	2425
12	Soybean-clean	30	563
13	Wheat	30	175
14	Woods	30	1235
15	Buildings-grass-trees	30	356
16	Stone-steel-towers	30	63
	Total	450	9799

TABLE II
STATISTICS OF THE PAVIAU DATASET

Class No.	Class Name	Training	Testing
1	Asphalt	30	6601
2	Meadows	30	18619
3	Gravel	30	2069
4	Trees	30	3034
5	Metal Sheets	30	1315
6	Bare Soil	30	4999
7	Bitumen	30	1300
8	Bricks	30	3652
9	Shadows	30	917
	Total	270	42506

TABLE III
STATISTICS OF THE HOUSTON DATASET

Class No.	Class Name	Training	Testing
1	Healthy grass	30	1221
2	Stressed grass	30	1224
3	Synthetic grass	30	667
4	Trees	30	1214
5	Soil	30	1212
6	Water	30	295
7	Residential	30	1238
8	Commercial	30	1214
9	Road	30	1222
10	Highway	30	1197
11	Railway	30	1205
12	Parking Lot1	30	1203
13	Parking Lot2	30	439
14	Tennis Court	30	398
15	Running Track	30	630
	Total	450	14579

A. Datasets

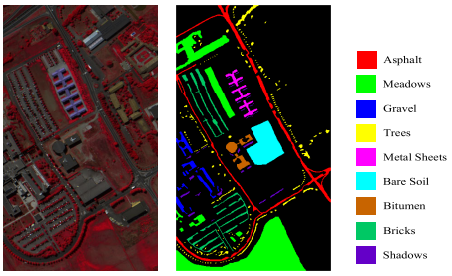
In these experiments, we evaluate the proposed method on three widely used HSIs. A false-color representation and the related ground truth for the three datasets are shown in Figs. 5–7. Besides, Tables I–III detail the statistics of the three datasets including the size of training and testing samples.

The first dataset is Indian Pines, acquired by the airborne visible/infrared imaging spectrometer (AVIRIS) sensor over Northwest Indian. This dataset contains 16 classes and has



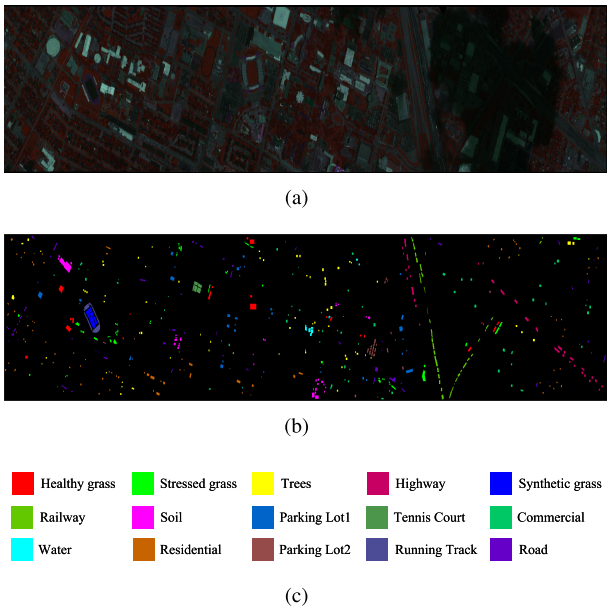
(a) (b) (c)

Fig. 5. (a) False-color representation of the Indian Pines dataset (bands 29, 19, and 9). (b) Ground truth for the Indian Pines dataset. (c) Color map.



(a) (b) (c)

Fig. 6. (a) False-color representation of the PaviaU dataset (bands 80, 60, and 35). (b) Ground truth for the PaviaU dataset. (c) Color map.



(a)

(b)

(c)

Fig. 7. (a) False-color representation of the Houston dataset (bands 101, 35, and 63). (b) Ground truth for the Houston dataset. (c) Color map.

145 × 145 pixels with 20 m/pixel spatial resolution. After removing 24 water absorption bands, there are 200 spectral bands left in Indian Pines covering the spectrum from 0.4 to 2.5 μm.

The second dataset is PaviaU, collected by the reflective optics system imaging spectrometer (ROSIS) sensor over the PaviaU, Italy. This dataset contains nine classes and has 610 × 340 pixels with 1.3 m/pixel spatial resolution. After removing 12 noisy bands, there are 103 spectral bands left in PaviaU covering the spectrum from 0.43 to 0.86 μm.

The third dataset is University of Houston (Houston), acquired by the compact airborne spectrographic imager (CASI) sensor over the University of Houston, USA. This dataset contains 15 classes and has 349 × 1905 pixels with 2.5 m/pixel spatial resolution. There are 144 spectral

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TABLE IV

CLASSIFICATION PERFORMANCE ON THE INDIAN PINES DATASET. BOLD: BEST; UNDERLINE: SECOND-BEST

ID	3DCNN	SSRN	A^2S^2K	WFCG	3DGAN	SSTN	GTFN	DGU	DRGCN	MSSGU	OGU	Proposed
1	96.87±5.03	**100.00±0.00**	80.11±8.63	81.23±3.97	97.32±2.35	98.93±2.83	**100.00±0.00**	**100.00±0.00**	97.51±3.06	**100.00±0.00**	**100.00±0.00**	**100.00±0.00**
2	70.90±4.85	89.55±3.68	87.64±3.87	86.54±3.05	78.46±4.09	84.56±3.93	89.92±0.78	68.06±8.61	88.26±5.23	92.19±4.26	94.84±2.63	**97.78±0.48**
3	65.70±6.09	83.11±4.06	85.96±5.46	81.04±2.93	85.47±3.25	85.16±5.44	91.71±0.52	55.16±5.41	90.55±1.71	**97.50±3.21**	96.12±3.14	96.16±0.46
4	**84.73±4.10**	**99.71±0.44**	**85.06±7.34**	**78.83±3.51**	**83.95±5.04**	**98.89±1.70**	99.03±1.41	88.74±3.71	95.45±5.24	99.75±0.32	99.95±0.28	**100.00±0.00**
5	86.35±8.02	**89.27±2.59**	92.08±3.18	97.26±0.85	85.39±5.58	91.05±2.69	95.39±0.20	76.21±8.55	93.37±1.38	95.95±1.94	97.77±2.36	**98.01±0.83**
6	92.68±4.88	**99.57±0.62**	95.46±1.87	98.43±0.92	88.75±1.47	97.16±1.45	96.65±1.32	97.99±1.02	95.42±1.81	98.12±0.86	98.17±0.73	99.42±0.86
7	96.92±3.76	**100.00±0.00**	51.58±10.63	73.08±13.86	62.47±11.69	**100.00±0.00**	94.47±3.25	**100.00±0.00**	**100.00±0.00**	**100.00±0.00**	**100.00±0.00**	**100.00±0.00**
8	**96.80±1.99**	**96.87±9.37**	**99.54±0.39**	**100.00±0.00**	95.02±1.36	99.47±0.23	99.05±0.21	94.91±0.13	99.73±0.21	**100.00±0.00**	**100.00±0.00**	**100.00±0.00**
9	**100.00±0.00**	**100.00±0.00**	35.64±6.09	**100.00±0.00**	80.06±6.28	99.43±1.38	**100.00±0.00**	**100.00±0.00**	**100.00±0.00**	**100.00±0.00**	**100.00±0.00**	**100.00±0.00**
10	78.57±5.26	90.34±3.67	66.35±6.84	80.03±2.66	80.19±4.92	88.41±3.91	95.80±1.32	68.89±4.12	88.95±5.24	95.08±4.43	94.02±2.76	**96.60±2.32**
11	64.39±5.27	79.90±4.81	**97.84±1.46**	94.60±3.02	88.21±2.35	74.82±5.45	90.18±0.64	62.34±6.72	88.46±5.35	91.36±2.95	93.02±1.80	94.84±2.56
12	**77.94±5.59**	94.72±2.50	84.46±7.93	80.09±7.35	75.47±6.85	95.53±2.28	89.21±2.09	69.51±6.05	93.64±2.44	96.98±1.49	95.91±4.39	**99.42±2.57**
13	99.54±0.66	99.77±0.52	84.83±3.97	98.19±1.70	85.54±5.36	99.65±0.43	99.26±0.28	99.48±1.03	**100.00±0.00**	**100.00±0.00**	**100.00±0.00**	99.42±0.24
14	86.38±12.11	98.06±1.64	98.67±0.64	97.37±6.12	92.17±2.81	97.42±1.56	99.10±0.57	93.56±2.63	97.82±3.79	99.12±0.36	98.76±2.85	**99.83±0.18**
15	73.51±8.23	75.11±4.42	88.09±5.98	**100.00±0.00**	85.25±4.86	98.36±2.61	99.15±0.36	56.57±4.29	98.87±1.26	98.85±2.50	99.85±0.51	**100.00±0.00**
16	98.73±1.18	99.52±0.72	84.96±4.61	**100.00±0.00**	73.42±6.59	97.98±3.12	99.34±0.73	**100.00±0.00**	**100.00±0.00**	99.84±0.46	**100.00±0.00**	**100.00±0.00**
OA	76.58±2.60	89.08±1.48	87.26±1.43	87.98±0.36	85.93±2.24	92.43±1.67	93.89±0.13	73.62±2.14	92.26±1.35	95.85±0.79	96.84±0.55	**97.19±0.75**
AA	85.62±1.33	93.46±0.88	80.04±1.37	86.02±1.53	83.62±1.48	93.05±1.16	96.28±0.28	83.23±1.27	95.55±0.64	97.79±0.37	98.31±0.29	**98.43±0.30**
Kappa	73.59±2.79	87.62±1.66	86.02±1.97	86.13±0.32	84.09±2.37	91.93±1.39	93.10±0.19	70.29±2.34	91.11±1.52	95.26±0.90	96.38±0.62	**96.78±0.86**

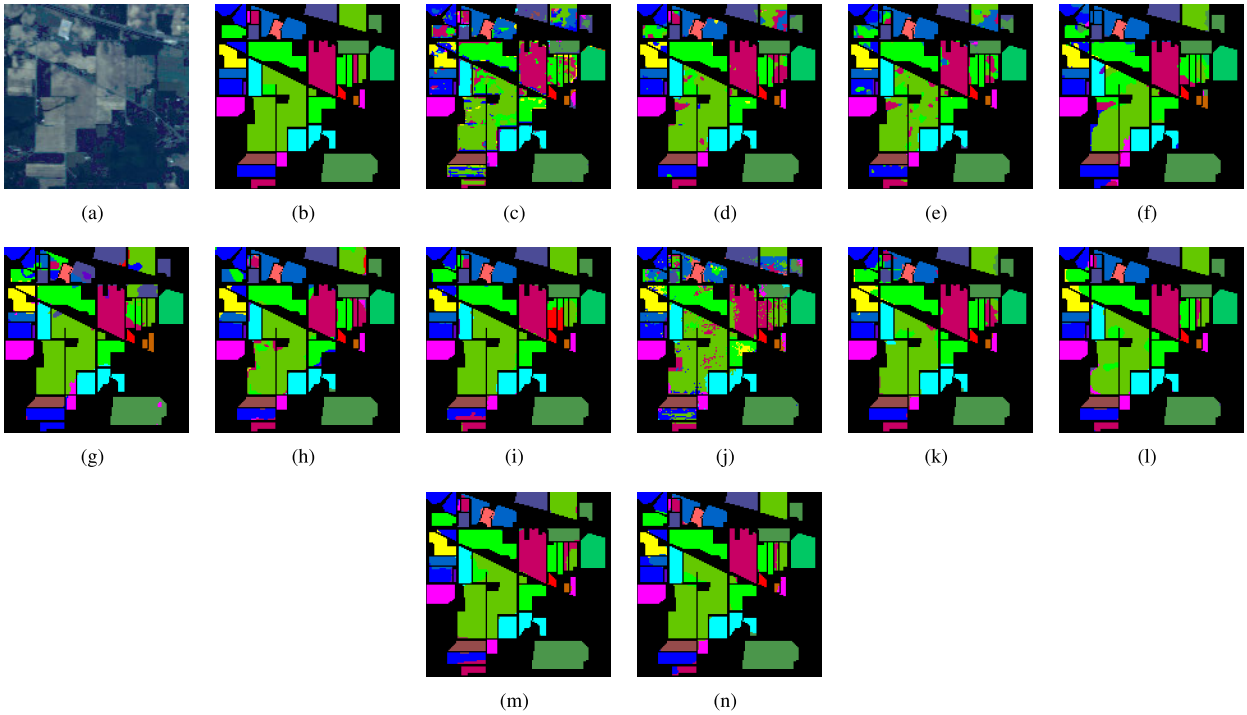


Fig. 8. Classification maps of different methods on the Indian Pines dataset. (a) False-color image (bands 29, 19, and 9). (b) Ground truth. (c) 3DCNN (OA = 76.58%). (d) SSRN (OA = 89.08%). (e) A^2S^2K (OA = 87.26%). (f) WFCG (OA = 87.98%). (g) 3DGAN (OA = 85.93%). (h) SSTN (OA = 92.43%). (i) GTFN (OA = 93.89%). (j) DGU (OA = 73.62%). (k) DRGCN (OA = 92.26%). (l) MSSGU (OA = 95.85%). (m) OGU (OA = 96.84%). (n) Proposed (OA = 97.19%).

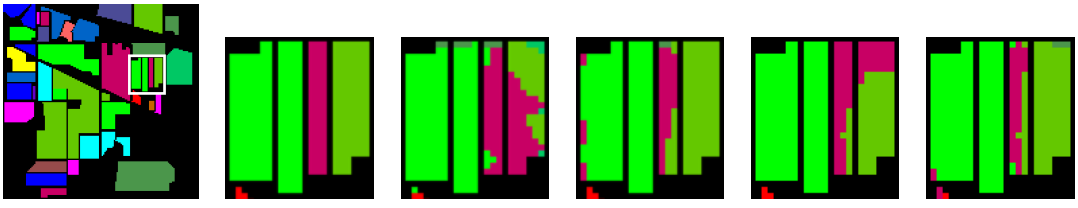


Fig. 9. Visualization results of typical areas [white rectangular frame in (a)] in the Indian Pines dataset with different methods. (a) Entire ground-truth map. (b) Ground-truth map. (c) DRGCN. (d) MSSGU. (e) OGU. (f) Proposed.

bands in the Houston dataset covering the spectrum from 0.38 to 1.05 μm.

B. Experimental Setup

In the experiments, the proposed method is implemented in PyTorch with Adam optimizer. The learning rate is 5e-4. The number of epochs is 200, 100, and 150 for Indian Pines, PaviaU, and Houston, respectively. In this article, for each dataset, we randomly choose 30 examples (i.e., pixels) per class for training or 15 examples if there are fewer than 30 examples in the corresponding class. To demonstrate the effectiveness of the proposed method, several state-of-the-art HSI classification methods are used for comparison, including the spatial–spectral CNN-based methods (3-D deep learning approach (3DCNN) [62],

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TABLE V

CLASSIFICATION PERFORMANCE ON THE PAVIAU DATASET. BOLD: BEST; UNDERLINE: SECOND-BEST

ID	3DCNN	SSRN	A^2S^2K	WFCG	3DGAN	SSTN	GTFN	DGU	DRGCN	MSSGU	OGU	Proposed
1	**79.42±6.58**	90.07±3.68	94.25±3.03	**98.19±0.52**	78.64±1.56	93.09±1.79	88.92±0.81	61.73±5.49	84.21±3.49	97.94±2.37	91.15±2.15	88.41±2.70
2	69.36±6.11	87.42±0.31	84.72±6.76	97.90±0.31	93.00±1.72	89.62±3.37	97.04±0.57	71.81±3.04	93.16±4.88	85.62±3.41	94.64±3.89	**98.36±1.06**
3	84.73±4.92	70.79±6.78	89.34±7.38	80.11±7.93	90.18±3.94	86.33±5.34	84.09±3.05	75.33±3.78	97.14±0.95	**100.00±0.00**	98.67±2.89	99.66±0.26
4	92.81±1.49	96.53±0.59	80.06±5.37	94.92±3.39	**98.62±1.49**	93.24±1.53	95.16±0.53	87.71±1.62	79.21±4.01	88.79±4.25	89.78±3.18	87.96±3.83
5	99.61±0.42	97.26±0.16	99.64±0.14	98.93±0.19	99.08±0.17	99.34±0.12	98.69±0.42	99.07±0.13	99.61±0.31	99.75±0.09	**99.94±0.09**	99.65±0.44
6	78.12±4.84	92.47±0.64	86.41±8.63	79.93±4.03	96.66±1.59	98.23±1.34	99.28±0.18	88.94±3.33	99.34±1.09	99.06±1.24	99.69±0.29	**100.00±0.00**
7	89.48±5.88	99.51±0.33	86.79±7.84	85.17±6.02	95.36±2.25	99.15±0.23	98.73±0.53	98.19±0.93	99.21±0.89	**100.00±0.00**	**100.00±0.00**	**100.00±0.00**
8	85.76±3.38	94.86±4.15	81.24±4.83	88.91±0.32	78.41±3.97	93.23±4.12	91.70±2.74	59.43±5.01	93.35±3.57	**99.13±0.85**	97.59±1.70	98.69±0.51
9	**99.45±0.49**	96.34±1.81	94.48±5.96	98.06±1.05	98.38±0.24	97.81±1.29	95.93±0.67	96.66±1.18	85.34±4.33	98.96±1.16	95.13±2.41	94.29±1.48
OA	77.99±6.05	89.42±1.67	85.31±1.64	92.79±2.34	90.52±2.67	93.42±2.43	95.04±0.19	74.71±1.69	91.93±1.12	93.49±1.63	95.66±1.64	**96.36±0.65**
AA	86.53±2.14	91.19±0.74	92.63±2.01	91.60±1.36	91.05±1.15	94.31±0.69	94.37±0.53	82.18±0.87	92.28±0.71	94.65±0.71	95.98±0.61	**96.33±0.44**
Kappa	72.55±6.46	86.29±0.98	81.49±1.97	90.75±2.86	88.59±2.09	90.78±1.91	93.18±0.36	68.41±1.97	89.46±1.63	90.08±1.96	94.27±2.07	**95.18±0.85**

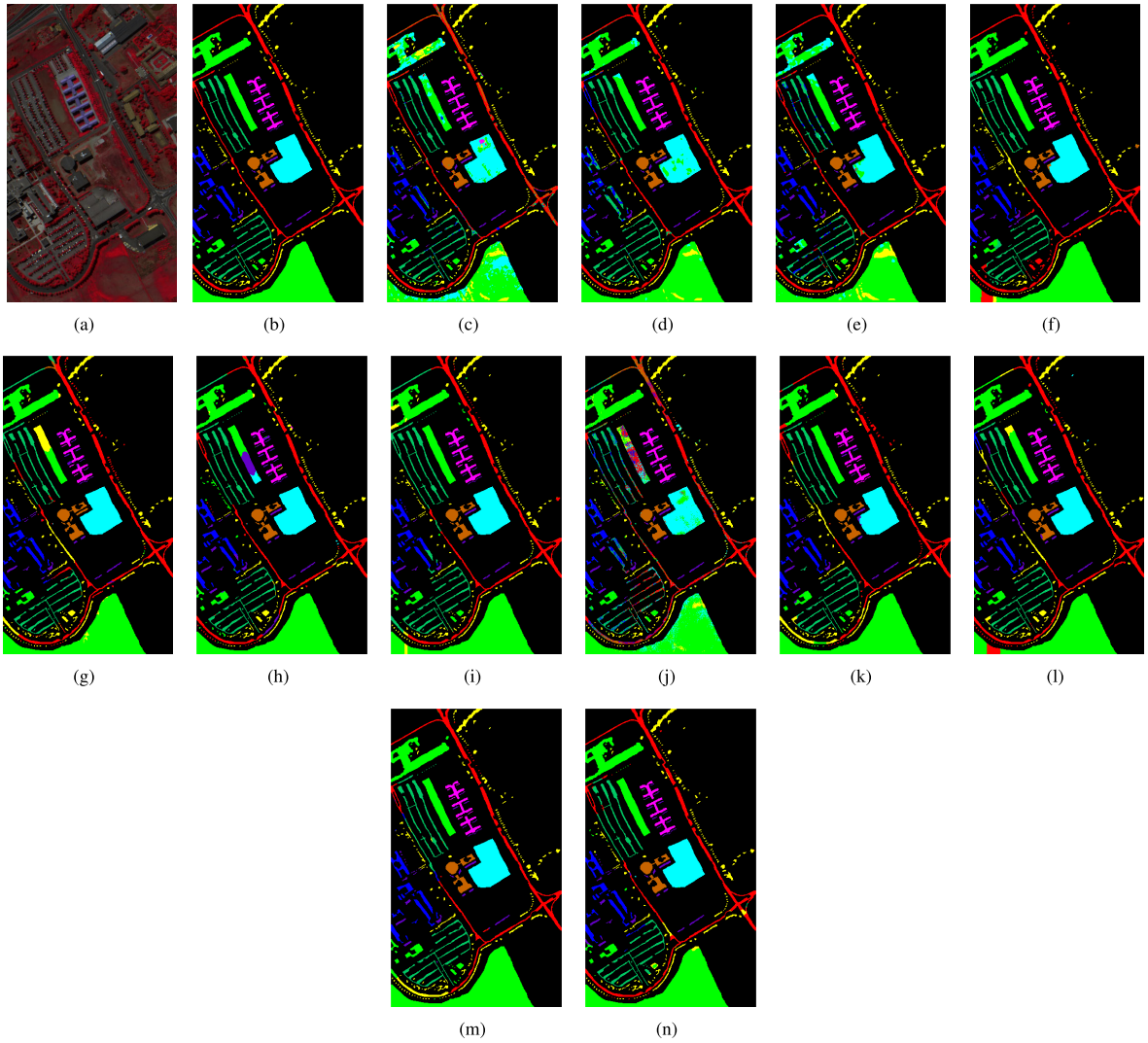


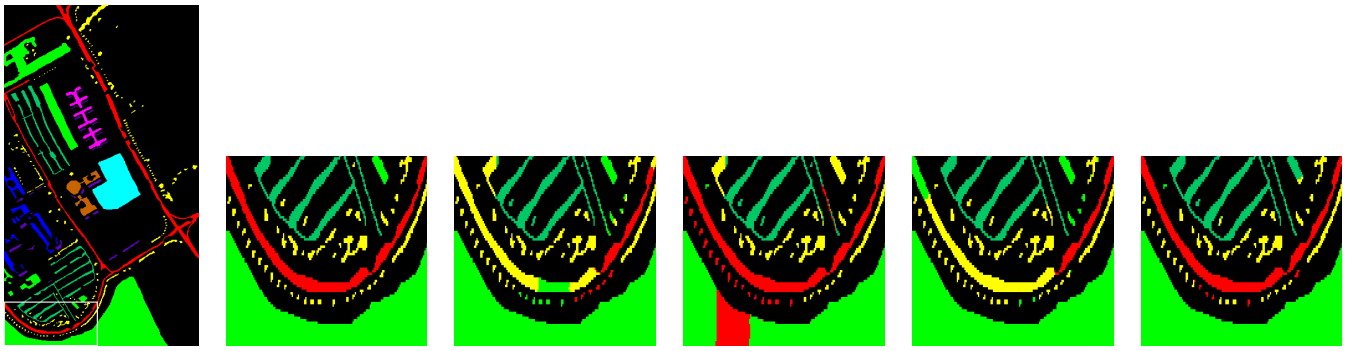
Fig. 10. Classification maps of different methods on the PaviaU dataset. (a) False-color image (bands 80, 60, and 35). (b) Ground truth. (c) 3DCNN (OA = 77.99%). (d) SSRN (OA = 89.42%). (e) A^2S^2K (OA = 85.31%). (f) WFCG (OA = 87.98%). (g) 3DGAN (OA = 90.52%). (h) SSTN (OA = 93.42%). (i) GTFN (OA = 95.04%). (j) DGU (OA = 74.71%). (k) DRGCN (OA = 91.93%). (l) MSSGU (OA = 93.49%). (m) OGU (OA = 95.66%). (n) proposed (OA = 96.36%).

spectral–spatial residual network (SSRN) [63], attention-based adaptive spectral–spatial kernel ResNet (A^2S^2K) [64], weighted feature fusion of CNN and graph attention network (WFCG) [65]), the GAN-based method (3DGAN [27]), the transformer-based methods (SSTN [66] and GTFN [67]), and the GCN-based methods (dual graph U-Nets (DGU) [50], dual residual GCN (DRGCN) [68], MSSGU [51], OGU [52]). The hyperparameters of the compared approaches are in agreement with the ones used in the original papers. All these methods are run ten times to obtain stable and reliable results on each HSI dataset. The AA and the related standard deviation are reported. All the experiments are executed on a computer with an AMD Epyc 7713 at 2-GHz × 2 with 2-TB RAM and Nvidia A100 GPU with 40-GB RAM.

C. Experimental Results

Eleven state-of-the-art deep-learning-based methods are used to compare with the proposed method on the Indian Pines, PaviaU, and Houston datasets, which are shown in Tables IV–VI. The best and the second-best classification results are highlighted in bold and underlined, respec-

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(a) (b) (c) (d) (e) (f)

Fig. 11. Visualization results of typical areas [white rectangular frame in (a)] in the PaviaU dataset with different methods. (a) Entire ground-truth map. (b) Ground-truth map. (c) DRGCN. (d) MSSGU. (e) OGU. (f) Proposed.

TABLE VI

CLASSIFICATION PERFORMANCE ON THE HOUSTON DATASET. BOLD: BEST; UNDERLINE: SECOND-BEST

ID	3DCNN	SSRN	A^2S^2K	WFCG	3DGAN	SSTN	GTFN	DGU	DRGCN	MSSGU	OGU	Proposed
1	94.02±6.21	**97.55±0.78**	95.43±3.16	95.04±3.56	83.81±4.17	97.14±2.06	60.48±4.03	81.21±3.93	82.94±7.26	93.45±6.05	92.46±5.07	95.49±1.54
2	94.05±5.81	90.62±3.56	**97.11±2.38**	83.59±5.36	82.04±3.41	92.38±1.03	90.69±3.76	89.97±4.80	93.45±4.69	93.90±4.58	88.97±3.36	90.84±2.10
3	97.96±1.11	97.58±1.21	99.06±0.87	92.83±4.12	92.19±1.55	99.05±0.31	91.75±1.49	90.33±0.33	98.88±0.11	**99.59±0.33**	98.65±3.01	98.90±0.56
4	94.99±3.07	90.45±4.53	98.12±2.63	78.56±4.85	89.35±3.26	92.46±2.68	81.05±2.05	81.92±2.66	89.76±3.95	**98.14±2.64**	95.63±2.89	92.50±2.99
5	97.45±2.46	93.98±2.76	90.55±2.31	89.47±2.05	90.11±2.06	94.84±3.73	92.83±3.87	91.73±0.24	97.34±2.11	99.12±1.90	98.26±3.34	**99.43±1.26**
6	89.42±5.32	86.02±5.93	97.31±3.74	94.03±3.08	80.34±4.25	99.41±3.19	87.64±2.51	88.24±2.63	98.71±0.84	99.52±0.66	94.33±2.14	**100.00±0.00**
7	80.14±10.21	84.99±3.62	93.18±4.02	85.42±3.21	71.43±5.87	70.15±7.53	88.32±4.08	64.62±0.51	88.43±4.63	**93.36±4.32**	88.85±2.91	91.68±5.40
8	58.60±11.81	91.21±5.94	**93.42±6.15**	88.35±6.09	22.96±10.34	68.77±6.32	74.84±5.60	58.67±7.82	81.13±3.20	86.34±5.87	85.42±0.44	83.69±4.24
9	75.94±4.84	89.04±3.19	92.78±3.58	90.23±4.02	68.15±5.34	91.43±4.71	83.08±5.80	46.86±6.81	**95.07±5.68**	88.72±5.49	81.91±1.48	93.20±3.16
10	51.76±22.24	94.68±3.05	85.63±3.16	89.32±1.35	60.14±8.53	89.36±5.19	84.85±5.34	62.94±5.81	98.33±3.44	94.99±7.50	98.21±1.55	**98.83±0.94**
11	65.25±21.30	94.12±0.98	91.05±2.74	82.09±3.06	68.41±4.33	79.70±6.37	92.18±5.16	73.08±5.07	97.11±3.45	95.52±4.62	97.01±3.24	**97.42±1.89**
12	76.20±12.84	95.21±1.63	90.33±4.83	73.13±6.82	57.44±3.97	72.51±5.09	65.70±5.04	70.15±4.35	91.63±2.99	90.93±5.13	87.02±2.19	**95.66±3.46**
13	74.92±11.03	91.85±4.39	96.08±3.47	88.24±1.72	38.73±6.15	**96.84±3.83**	87.51±3.68	75.91±5.76	85.06±5.83	92.75±5.75	89.97±1.51	91.57±1.28
14	97.86±1.49	**100.00±0.00**	**100.00±0.00**	96.97±2.06	90.47±4.07	**100.00±0.00**	90.92±1.33	99.76±0.22	**100.00±0.00**	**100.00±0.00**	**100.00±0.00**	**100.00±0.00**
15	98.85±0.85	96.35±2.01	98.25±1.34	95.80±3.92	93.21±3.45	98.57±1.54	91.04±1.95	97.32±1.94	96.03±3.45	99.16±0.06	99.36±1.61	**99.52±0.07**
OA	81.25±2.99	91.57±1.58	93.28±1.63	90.51±2.71	73.51±1.61	92.24±1.19	80.09±1.97	72.69±1.18	92.22±1.51	94.29±1.03	93.31±1.01	**94.54±0.98**
AA	83.16±2.27	92.81±0.97	94.95±0.87	92.29±2.15	74.09±1.04	93.31±1.07	82.15±1.21	75.72±0.82	92.76±1.45	95.09±0.85	94.19±1.88	**95.21±0.81**
Kappa	79.73±3.22	91.08±1.83	93.12±1.24	90.63±2.81	72.91±1.93	91.56±1.42	78.81±1.73	70.89±1.28	91.54±1.64	93.82±1.11	92.77±1.26	**94.09±1.06**

tively. The corresponding classification maps on the three datasets are shown in Figs. 8, 10, and 12, respectively. Close-ups of typical areas for the three datasets provided by the different methods are shown in Figs. 9, 11, and 13, respectively.

For the Indian Pines and PaviaU datasets, the proposed method obtains much better classification results than the competitors. More specifically, the OA of the proposed method outperforms the second-best method (OGU) on the two datasets by 0.35% and 0.7%, respectively. Besides, our method achieves the smallest standard deviation compared with the other methods in terms of OA, AA, and Kappa, respectively, thus demonstrating its stability. Moreover, it is worth noting that our method ranks first for 12 classes on the Indian Pines dataset. Since this HSI has many ground objects with different sizes, the topology-feature pooling layer plays a crucial role in extracting the various spatial topologies and discriminative features. Compared with transform-based methods, the OA of the proposed method outperforms GTFN on the two datasets by 3.3% and 1.32%, respectively. Thanks to the cross-attention mechanism, the proposed method can learn complementary multiresolution features to find more useful information for various ground objects, thereby obtaining the best classification performance. The classification maps of the Indian Pines and PaviaU datasets are shown in Figs. 8 and 10, respectively. The false-color images and ground-truth maps are also presented. We can note that the classification maps of the proposed method are smoother than the ones of the compared approaches with less pepper-salt noise. It is worth noting that the proposed method yields a better performance on long and narrow regions [e.g., "Asphalt" (ID 1)] on the PaviaU dataset.

For the Houston dataset, the proposed method achieves the best classification results compared with the other methods. As can be seen from Fig. 12(a), there are many small-scale land covers with a staggered distribution in the Houston dataset. Nevertheless, the proposed method achieves an accuracy of 94.54%, and it ranks first in the classification of seven classes, which proves the capacity of our method to deal with complex scenarios. Besides, compared with the CNN-based methods, we can see that the GCN-based methods (i.e., DRGCN and MSSGU) obtain higher classification results because they can extract the region boundaries during the training process. It is worth noting that our method achieves better accuracies on some small regions [e.g., "Railway" (ID 11) and "Parking Lot1" (ID 12)], which demonstrates its effectiveness in modeling the spatial topology of HSIs. From the classification map in Fig. 12, the proposed method obtains optimal visual results compared with the other approaches.

D. Impact of the Number of Training Samples

The size of the training samples has a significant effect on classification accuracy. Most of the HSI classification methods demonstrate that they can obtain good classification performance using enough training samples. However, the scarcity of labeled samples (often attributed to the high cost of human resources to generate them) poses a significant challenge. Hence, HSI classification performance under limited training samples is another crucial evaluation criterion. In this section, 5, 10, 15, 20, 25, and 30 training samples are selected from each class to evaluate the performance of the proposed method. The OAs obtained by some representative

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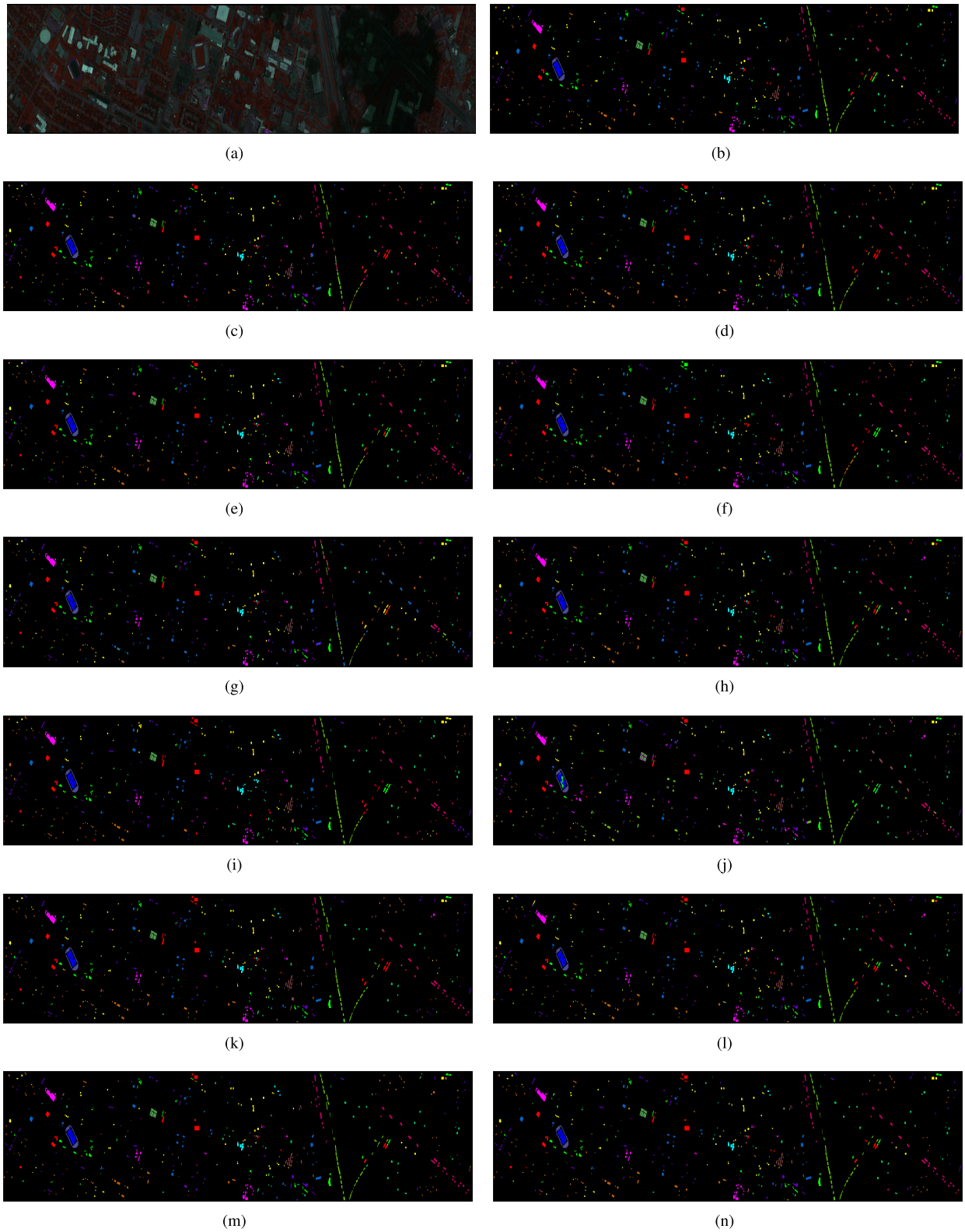


Fig. 12. Classification maps of different methods on the Houston dataset. (a) False-color image (bands 101, 35, and 63). (b) Ground truth. (c) 3DCNN (OA = 81.25%). (d) SSRN (OA = 91.57%). (e) A^2S^2K (OA = 93.28%). (f) WFCG (OA = 90.51%). (g) 3DGAN (OA = 73.51%). (h) SSTN (OA = 92.24%). (i) GTFN (OA = 80.09%). (g) DGU (OA = 72.69%). (k) DRGCN (OA = 92.22%). (l) MSSGU (OA = 94.29%). (m) OGU (OA = 93.31%). (n) Proposed (OA = 94.54%).

methods on the three datasets are reported in Fig. 14. It can be observed that our method obtains better classification performance than the other compared techniques. Besides, it is worth to be remarked that the proposed method achieves a higher accuracy than the other methods on the three datasets. Since MSSGU can extract multiresolution features from a hierarchical segmentation graph, it obtains a classification performance close to our approach for the Houston dataset.

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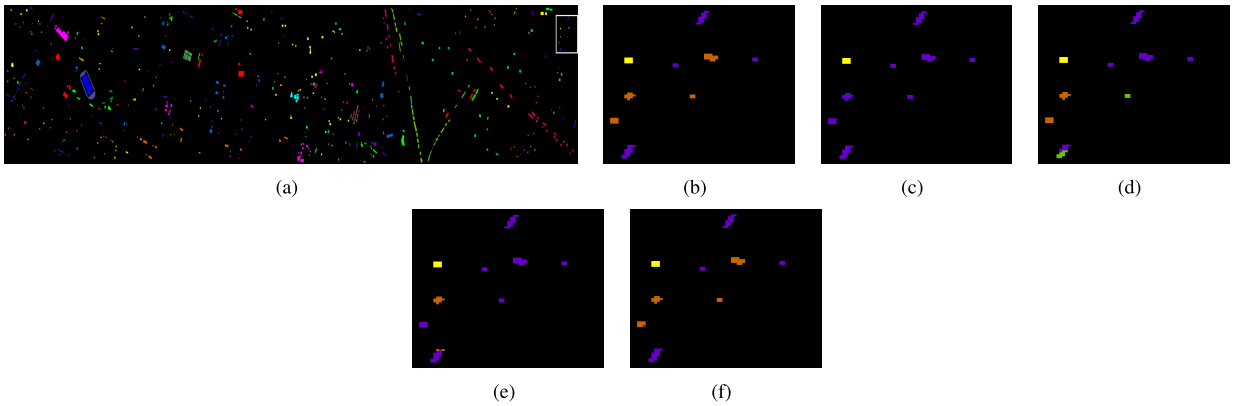


Fig. 13. Visualization results of typical areas [white rectangular frame in (a)] in the Houston dataset with different methods. (a) Entire ground-truth map. (b) Ground-truth map. (c) DRGCN. (d) MSSGU. (e) OGU. (f) Proposed.

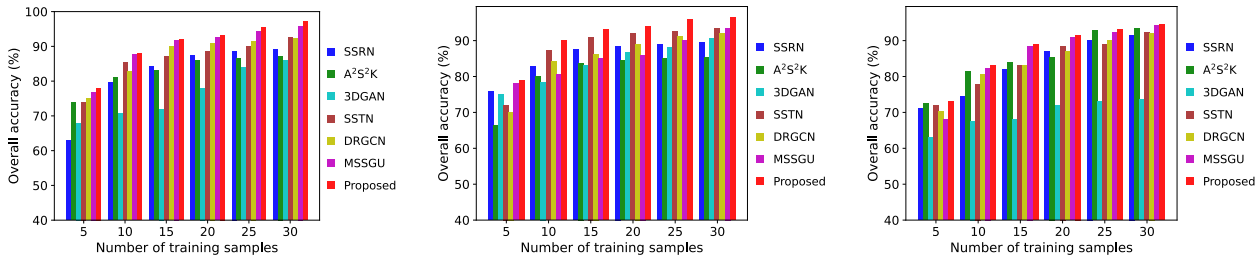


Fig. 14. OAs obtained by some representative methods varying the numbers of training samples per class. (a) Indian Pines. (b) PaviaU, and (c) Houston.

E. Impact of Network Depth and Hidden Kernel Size

In this section, we analyze the impact of network depth and hidden kernel size of GCN layers on the three datasets. Finding appropriate parameters is crucial for neural network models to alleviate overfitting issues, thus enhancing their robustness and generalization ability. The OAs of the proposed method under different network depths (i.e., the number of blocks containing TFAP layers) and hidden kernel sizes on the Indian Pines, PaviaU, and Houston datasets are reported in Tables VII–IX, respectively. The results for different network depths in Tables VII–IX are obtained optimizing the node sampling ratio k in the set $\{0.9, 0.8, 0.7, 0.6\}$. We can observe that the proposed method obtains satisfactory classification results for network depths equal to 2 and 3. For Indian Pines, the network depth and hidden kernel size of GCN layers are set to 2 and 50, respectively, when k is set to $\{0.8, 0.7\}$. On the PaviaU dataset, the best OA values for the network depth and the hidden kernel size of GCN layers are set to 2 and 40, respectively, when k is set to $\{0.9, 0.8\}$. Owing to the complex distribution of land covers in the Houston dataset, the deepening of the network can extract more discriminative features from the input HSI, and thus the best OA value is obtained when the network depth and hidden kernel size of GCN layers are set to 3 and 40, respectively, with k equal to $\{0.9, 0.8, 0.7\}$.

F. Impact of Segmentation Scale

The segmentation scale, δ, fixes the size of the initial graph. Hence, we analyzed its effect on our method. Specifically, we set δ with values in the range from 100 to 500, considering a step size equal to 100. The performance of our method

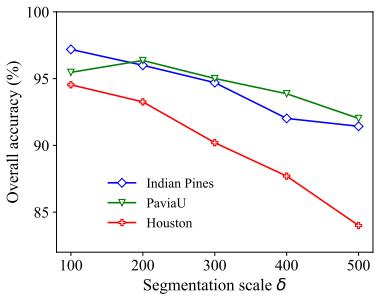


Fig. 15. Performance of the proposed method varying the segmentation scale on the three datasets.

varying δ on the three datasets is shown in Fig. 15. As can be observed, with the increase in the segmentation scale, the classification performance of the Houston dataset shows more fluctuations than the ones for the Indian Pines and PaviaU datasets. The reason is that the Houston dataset contains many small and irregularly shaped land covers. The larger the segmentation scale, the more the pixels of different categories belong to the same superpixel. As a result, the classification performance of the proposed model decreases when the segmentation scale increases for the Houston dataset. On the contrary, the Indian Pines and PaviaU datasets contain many large-scale land covers, and thus, they are insensitive to the segmentation scale. In this article, the segmentation scale, δ, is set to 200 for PaviaU and 100 for Indian Pines and Houston.

G. Ablation Studies

In this section, we analyze the contributions of four crucial components (i.e., topology structure learning, node

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TABLE VII

PERFORMANCE (OA) OF THE PROPOSED METHOD ON THE INDIAN PINES DATASET UNDER DIFFERENT NETWORK DEPTHS AND HIDDEN KERNELS SIZES FOR THE GCN LAYERS. BOLD: BEST; UNDERLINE: SECOND-BEST

Hidden kernel size \ Network depth	1	2	3	4
20	94.73±0.62	94.51±0.63	94.32±0.69	82.91±1.45
30	95.84±0.25	95.28±0.68	95.41±0.50	88.34±1.65
40	95.93±0.73	96.16±0.53	96.22±0.89	90.75±1.36
50	96.37±0.79	**97.19±0.71**	96.43±0.75	80.39±2.13
60	96.31±0.55	96.30±0.44	96.27±0.69	76.89±3.51

TABLE VIII

PERFORMANCE (OA) OF THE PROPOSED METHOD ON THE PaviaU DATASET UNDER DIFFERENT NETWORK DEPTHS AND HIDDEN KERNELS SIZES FOR THE GCN LAYERS. BOLD: BEST; UNDERLINE: SECOND-BEST

Hidden kernel size \ Network depth	1	2	3	4
20	95.02±0.94	95.55±0.92	95.60±1.18	95.80±1.25
30	95.99±0.81	96.01±0.47	96.24±0.93	95.69±0.84
40	96.06±0.77	**96.36±0.65**	95.91±0.55	95.79±0.69
50	96.12±0.53	96.29±0.31	95.97±0.91	96.02±0.78
60	96.09±0.58	95.85±0.72	95.89±0.54	95.96±0.99

TABLE IX

PERFORMANCE (OA) OF THE PROPOSED METHOD ON THE HOUSTON DATASET UNDER DIFFERENT NETWORK DEPTHS AND HIDDEN KERNELS SIZES FOR THE GCN LAYERS. BOLD: BEST; UNDERLINE: SECOND-BEST

Hidden kernel size \ Network depth	1	2	3	4
20	90.71±1.41	91.73±1.32	92.70±1.04	90.52±1.25
30	91.31±1.45	93.15±1.18	93.18±1.19	93.14±1.08
40	92.35±1.39	93.50±0.68	**94.54±0.98**	93.36±0.74
50	93.15±1.36	93.44±0.71	93.33±0.77	93.45±0.80
60	92.85±1.02	93.25±0.41	92.72±0.52	92.86±0.90

TABLE X

ABLATION STUDIES FOR THE PROPOSED METHOD. BOLD: BEST; UNDERLINE: SECOND-BEST

Datasets	Methods	OA	AA	Kappa
Indian Pines	w/o-topology structure learning	95.31±1.15	97.03±0.38	94.63±1.31
	w/o-node feature learning	95.72±1.13	97.52±0.57	95.09±1.28
	w/o-cross attention module	96.23±1.61	97.82±0.63	95.18±1.82
	Proposed-simple pooling	93.76±6.08	94.64±4.41	92.75±6.79
	Proposed	**97.19±0.75**	**98.43±0.30**	**96.78±0.86**
PaviaU	w/o-topology structure learning	95.26±0.94	95.35±0.83	93.74±0.51
	w/o-node feature learning	95.13±0.79	96.01±0.43	93.59±1.38
	w/o-cross attention module	95.96±1.64	96.21±0.86	94.67±1.03
	Proposed-simple pooling	91.38±2.78	93.37±1.78	88.78±3.54
	Proposed	**96.36±0.65**	**96.33±0.44**	**95.18±0.85**
Houston	w/o-topology structure learning	92.49±1.09	93.91±0.75	91.88±0.97
	w/o-node feature learning	93.13±0.64	94.34±0.73	93.05±0.85
	w/o-cross attention module	93.64±0.25	94.51±0.24	93.35±0.68
	Proposed-simple pooling	92.04±2.01	93.22±1.93	91.96±1.07
	Proposed	**94.54±0.98**	**95.21±0.81**	**94.09±1.06**

TABLE XI

COMPARISON OF TRAINING TIME, TESTING TIME, RUNNING TIME, AND NUMBER OF PARAMETERS FOR THE COMPARED METHODS ON THE THREE CONSIDERED DATASETS. BOLD: BEST; UNDERLINE: SECOND-BEST

Dataset	Time/Complexity	3DCNN	SSRN	A^2S^2K	3DGAN	SSTN	DGU	DRGCN	MSSGU	OGU	Proposed
Indian Pines	Train (s)	22.64	209.47	284.20	172.86	473.22	860.13	17.15	22.18	12.17	**7.55**
	Test (s)	0.56	4.41	2.97	2.54	2.96	198.81	3.02	2.03	0.28	**0.01**
	Run (s)	23.20	213.88	287.17	175.40	476.18	1058.94	20.17	24.21	12.45	**7.56**
	Params (K)	46.11	364.19	370.79	10460.3	**20.98**	142.42	87.82	133.05	143.82	122.92
PaviaU	Train (s)	12.24	77.97	94.18	205.40	864.53	743.26	179.08	51.42	268.09	**8.70**
	Test (s)	3.66	46.68	31.68	3.04	10.14	232.08	21.14	18.18	0.82	**0.62**
	Run (s)	15.90	124.65	125.86	208.44	874.67	975.34	200.22	69.60	268.91	**9.32**
	Params (K)	35.53	216.56	220.75	10460.2	**16.21**	100.06	87.39	119.53	84.78	78.85
Houston	Train (s)	**21.45**	160.79	209.41	197.41	196.74	1030.76	779.42	183.72	134.81	77.97
	Test (s)	15.35	151.79	115.14	124.52	70.37	760.73	59.44	58.37	0.08	**0.02**
	Run (s)	**36.80**	312.58	324.55	321.93	267.11	1791.49	838.86	242.09	134.89	77.99
	Params (K)	**41.28**	278.19	283.37	10461.09	72.06	104.37	87.75	125.64	81.54	84.14

feature learning, CAM, and pooling) on three datasets, where the training sample size is the same as in Section III-B. We report the OA, AA, and kappa coefficients obtained by the proposed method and the methods coming from

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the omission of one of the first three above-mentioned components and the change in the pooling strategy for the last component. More specifically, "w/o-topology structure learning," "w/o-node feature learning," and "w/o- CAM" indicate the techniques obtained by removing the topology structure learning, the node feature learning, and the CAM, respectively. Instead, "proposed-simple pooling" refers to use the same pooling function as in [55]. Table X reports the classification performance of the above-mentioned approaches on the three datasets, where the best and second-best values are highlighted in bold and underlined, respectively. It is clear that the classification performance gets worse when one among the topology structure learning, the node feature learning, and the CAMs is removed. Thus, this analysis reveals that each part of the proposed approach plays a crucial role in enhancing the classification performance. Finally, comparing the proposed pooling strategy with "proposed-simple pooling," we can easily note the improvement in HSI classification performance by adopting our pooling approach.

H. Running Time and Model Complexity

Table XI reports the performance parameters (i.e., training times, testing times, running times, and the number of parameters) of the different methods on the three considered datasets. It is clear that the proposed method obtains a comparable running time, thanks to the pooling layer, which reduces the number of graph nodes during the training process. Moreover, the testing time of the proposed method is also limited. It is worth noting that the CNN-based methods (i.e., SSRN, and A^2S^2K) use more parameters than the proposed method. Besides, compared with the other GCN-based methods, our model requires a similar number of parameters. However, it leverages on pooling layers to build a deep network, enabling the extraction of more discriminative spectral–spatial features.

IV. Conclusion

In this article, a graph U-Net with TFAP has been proposed for HSI classification. Unlike other GCN-based methods used for HSI classification that consider node features to select important nodes to obtain the pooled graph topology, we simultaneously considered the graph's topology structure and feature information during the pooling process. Besides, to allow nodes after pooling to contain more graph information, our method aggregated unselected node information before discarding nodes, which can ensure that significant features are retained to a certain extent. Moreover, we introduced a CAM to fuse different stage features, thus filtering out useless (or noisy) features. The experimental results on three HSI datasets demonstrated the superior performance of the proposed method with respect to some state-of-the-art approaches.

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